What Drives Next-Occupation Prediction?

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Abstract

Predicting a person's next occupation can support career guidance, re-skilling recommendations, public employment services, and labor-market analysis. We formulate next-occupation prediction as a ranking problem in which an observed career history is used to retrieve the most plausible next occupation from a standardized occupational vocabulary. We evaluate semantic retrieval, task-adapted representation learning, supervised career-transition signals, occupational knowledge generated offline by a large language model (LLM), and query-time LLM reasoning on JobHop v2, a large-scale public dataset of real career trajectories extracted from anonymized resumes and normalized to the European Skills, Competences, Qualifications and Occupations (ESCO) taxonomy. On the full test split, the strongest transition-aware model reaches a Mean Reciprocal Rank at 10 (MRR@10) of 0.155, while transition-only ranking already reaches 0.133, highlighting the strength of train-derived career-transition information. Frozen occupational Wiki knowledge is weak in isolation but provides a small statistically significant improvement when fused with transition evidence. In contrast, query-time Qwen reasoning provides no reliable additional gain: direct candidate selection significantly degrades ranking quality, while routing produces a change that is statistically indistinguishable from zero. Overall, our results show that supervised transition structure is a strong signal for next-occupation prediction, whereas LLM-based knowledge is most useful as a complementary offline source rather than as an automatically beneficial query-time reasoning component.

CCS Concepts

• **Information systems** → **Information retrieval**; • **Computing methodologies** → **Natural language processing**.

Keywords

Next-Occupation Prediction, Career Trajectories, Information Retrieval, ESCO, Large Language Models, Knowledge Augmentation

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1 Introduction

Career trajectories describe how workers move between occupations, accumulate skills, and respond to changes in the labor market over time. Predicting the next occupation from an observed career history is therefore relevant to career guidance, re-skilling recommendations, public employment services, and labor-market analysis. However, next-occupation prediction is not simply a semantic-matching problem. A plausible next occupation may be semantically related to previous jobs, empirically common after the current occupation, or supported by broader knowledge about occupational roles and transferable skills. These signals overlap, but they need not rank candidates in the same way.

We formulate next-occupation prediction as a retrieval problem. The observed career prefix forms the query, standardized occupations form the candidate documents, and the person's actual next occupation is the single relevant item. This formulation allows semantic retrievers, transition-based predictors, frozen occupational knowledge, and LLM-assisted ranking methods to be evaluated under a common ranking protocol.

A central question is how much prediction is driven by semantic similarity and how much is explained by transition regularities observed directly in historical career sequences. Prior work has shown that longitudinal career information is useful for person–job modeling [5] and that representations grounded in the European Skills, Competences, Qualifications and Occupations (ESCO) taxonomy can support job recommendation [14]. At the same time, career transitions provide a direct task-specific signal that may capture regularities not expressed by generic semantic similarity.

A second question concerns knowledge augmentation. Large language models (LLMs) have increasingly been used as reasoning components in recruitment and recommendation settings [19]. They can also be used offline to construct richer entity representations without invoking a generative model for every query. We therefore distinguish between frozen occupational knowledge generated once before evaluation and query-time LLM reasoning used for routing or candidate selection.

Rather than proposing a new sequence architecture, our study focuses on a diagnostic question: which sources of information actually drive next-occupation ranking performance, and when do additional knowledge or LLM components provide measurable value?

Our contributions are as follows:

- **Unified retrieval evaluation.** We compare dense, task-adapted, and late-interaction retrievers under the same next-occupation ranking formulation and test split.
- **Transition and error analysis.** We quantify the contribution of supervised career-transition information and complement aggregate results with paired significance tests and error analysis across return moves, seen and unseen transitions, career-history length, and target occupation frequency.

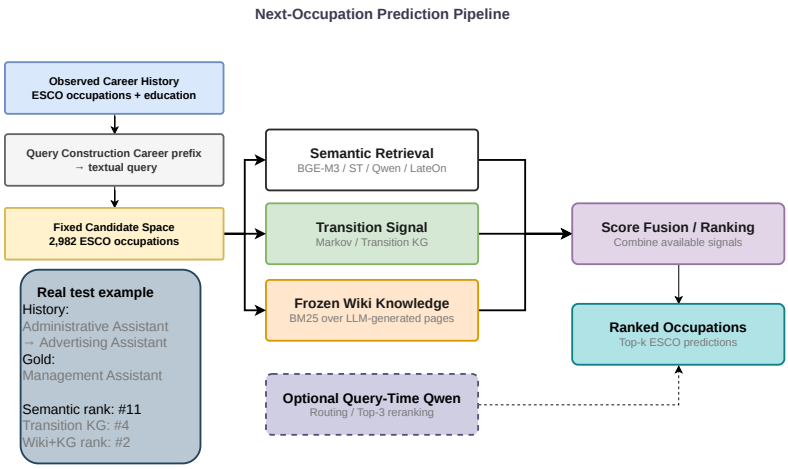


Figure 1: Overview of the next-occupation prediction pipeline. An observed career history is converted into a retrieval query over a fixed ESCO candidate space. Semantic retrieval, train-derived transition signals, and frozen occupational Wiki knowledge provide complementary ranking evidence that can be combined before final prediction. Query-time Qwen reasoning is evaluated separately as an optional downstream ablation. The micro-example shows a real non-return test instance in which frozen Wiki evidence improves the gold occupation's rank when fused with transition statistics.

- **Knowledge and LLM diagnostics.** We distinguish offline LLM-generated occupational knowledge from query-time LLM reasoning and measure their respective contributions beyond deterministic transition-based ranking.

2 Related Work

Career trajectories and occupational modeling. Career-path prediction aims to infer plausible future occupations from an individual's previous work history. The original JobHop dataset [8] provides large-scale career trajectories extracted from resumes, while JobHop v2 [6] extends the resource with a redesigned extraction pipeline and richer annotations, including quarter-level temporal information and normalized educational attainment. Both releases map work experiences to ESCO occupations, enabling evaluation over a standardized occupational space.

Several studies have investigated representations and models for career progression. Decorte et al. [2] formulate career-path prediction from textual work histories and compare text- and skill-based representations over ESCO occupations. KARRIEREWEGE [15] substantially expands the scale of public career-path data with more than 500,000 career paths linked to ESCO. Its KARRIEREWEGE+ variant augments the structured trajectories with synthetic job titles and descriptions to support prediction from free-text career inputs. Senger et al. [16] further study realistic career-path prediction across DECORTE and KARRIEREWEGE+, comparing conventional, neural, and LLM-based approaches.

For our evaluation, JobHop v2 is particularly suitable because it combines large-scale real resume-derived career trajectories with standardized ESCO occupation identifiers, quarter-level temporal information, and normalized educational attainment. This allows next-occupation ranking to be evaluated over a fixed occupational vocabulary without relying on synthetically generated career descriptions.

STEP [7] is particularly close to our setting because it directly evaluates next-job prediction on JobHop and JobHop v2. It combines temporal modeling, educational information, and learned occupation representations and compares against retrieval, Markov, linear, multilayer-perceptron, and recurrent baselines. Our goal is complementary rather than architectural: instead of introducing another sequence model, we diagnose the relative contributions of semantic retrieval, train-derived transition statistics, frozen occupational knowledge, and query-time LLM reasoning under a common ranking formulation.

Related work also learns occupational representations for recommendation. CareerBERT [14] embeds resumes and ESCO occupations in a shared representation space, while JobBERT [3] models job titles through their associated skills. Gong et al. [5] show that explicit career-path information can improve person–job fit, reinforcing the importance of longitudinal signals beyond isolated semantic matching.

Dense and late-interaction retrieval. Dense bi-encoder architectures provide an efficient framework for semantic ranking. Sentence-BERT [12] established efficient sentence-level embedding retrieval, while more recent retrieval architectures include BGE-M3 [1], Qwen3-Embedding-8B [23], and LateOn [17].

Recruitment-specific approaches such as ConFit and ConFit V2 [20, 21] formulate resume–job matching as dense retrieval and improve representation learning through contrastive learning, data augmentation, and hard-negative strategies. Our setting differs from conventional resume–vacancy matching because the query is

a longitudinal career prefix and the target is a standardized next occupation rather than a specific job posting. This makes it possible to study semantic retrieval together with transition signals derived directly from historical occupation sequences.

Knowledge-enhanced retrieval and occupational knowledge. ESCO [4] provides a standardized multilingual taxonomy of occupations, skills, competences, and qualifications. Taxonomy-aware models such as ESCOXLM-R [22] demonstrate that structured occupational information can improve representation learning for labor-market NLP.

Our frozen occupational Wiki is motivated by recent work on knowledge-enhanced and agent-oriented retrieval. LLM-Wiki [10] proposes organizing external knowledge into structured Wiki pages and exposing search, reading, link traversal, and self-correction operations to an LLM agent. Our use of this idea is deliberately narrower. We adopt only the offline entity-page construction principle: one static page is generated for each ESCO occupation from its identifier, title, and description, and the resulting collection is frozen before evaluation. We do not reproduce the full LLM-Wiki agentic search/read/link loop or its persistent self-correction mechanism.

LLMs and agentic career recommendation. LLMs have increasingly been incorporated into recruitment and career recommendation systems. GLRec [19] explores LLM reasoning over graph-structured recruitment information, showing how generative models can interact with structured job-domain knowledge. More recently, AdaptJobRec [18] introduces an agentic conversational career recommendation system that combines autonomous tool selection, memory processing, task decomposition, and personalized recommendation components.

These systems motivate the question of whether additional online LLM reasoning is useful once strong deterministic retrieval and transition signals are already available. Accordingly, our experiments distinguish between two uses of an LLM: offline knowledge construction, where occupational pages are generated once and frozen, and query-time reasoning, where the model is invoked for routing or candidate selection during prediction.

3 Benchmarking Framework

This section defines the task and the four complementary information settings considered in our evaluation.

Our study addresses four research questions (RQs):

- **RQ1:** How do modern retrieval architectures perform for next-occupation prediction?
- **RQ2:** How effective are task-adapted retrievers when combined with supervised occupation-transition signals?
- **RQ3:** Does frozen LLM-generated occupational knowledge provide additional value beyond supervised transition statistics?
- **RQ4:** Does query-time LLM reasoning improve ranking beyond the deterministic retrieval and transition pipeline?

3.1 Task Formulation

Let a career trajectory be an ordered sequence

$$T = (o_1, o_2, \ldots, o_N)$$

where N is the trajectory length and each $o_i \in O$ denotes an occupation from the candidate vocabulary O. Given the observed prefix

$$H = (o_1, o_2, \ldots, o_{N-1})$$

the objective is to rank the true next occupation o_N as highly as possible among the candidates in O.

We cast this problem as retrieval. The query contains previous occupations in the career history, their ESCO codes, and available education information. Each candidate document corresponds to one standardized occupation and contains its code, title, and ESCO description.

3.2 JobHop v2

We select JobHop v2 [6] because it provides large-scale real career trajectories normalized to a common ESCO occupation vocabulary, together with temporal and educational information suitable for controlled next-occupation evaluation. The dataset contains 355,315 raw career trajectories: 284,251 training trajectories, 35,531 validation trajectories, and 35,533 test trajectories.

Unknown or invalid ESCO codes are removed, jobs are ordered chronologically, consecutive duplicate occupations are collapsed, and only trajectories with at least two valid occupations after preprocessing are retained. The candidate vocabulary is derived exclusively from retained training trajectories.

After preprocessing, the formal experimental data contain 256,593 training trajectories, 32,123 validation trajectories, and 32,089 test trajectories. Each retained trajectory contributes exactly one last-step prediction instance: the prefix $(o_1, \ldots, o_{N-1})$ forms the query and the final occupation o_N is the single gold target. Therefore, the numbers of retained trajectories and prediction instances are identical for validation and test. The train-derived candidate vocabulary contains 2,982 occupations. All final validation and test targets occur in this candidate vocabulary, so the target-vocabulary check removes zero additional evaluation instances.

Training trajectories are used for two purposes. First, they provide supervision for task-specific retriever training. Second, consecutive occupation pairs are used to estimate empirical career-transition statistics.

3.3 Retriever-Only Ranking

The first experimental stage isolates retrieval from transition fusion and query-time LLM reasoning. Each retriever directly ranks all 2,982 candidate occupation documents.

We evaluate BGE-M3, a task-adapted Sentence Transformer, a task-adapted Qwen3-Embedding-8B model, and LateOn. These models represent general-purpose dense retrieval, task-adapted bi-encoder retrieval, LLM-based embeddings, and late-interaction retrieval, respectively.

This experiment establishes the standalone retrieval comparison used to answer RQ1.

3.4 Task-Specific Training and Transition Fusion

The task-adapted Sentence Transformer is optimized contrastively using in-batch negatives, while Qwen3-Embedding-8B is adapted

Preprocessing stage	Train	Validation	Test
Raw trajectories	284,251	35,531	35,533
Trajectories with at least one valid code	281,982	35,262	35,249
Length ≥ 2 before duplicate collapse	263,697	32,976	32,988
Length ≥ 2 after duplicate collapse	256,593	32,123	32,089
Target in train-derived vocabulary	–	32,123	32,089
Final prediction instances	256,593	32,123	32,089

Table 1: JobHop v2 preprocessing counts. Removing unknown or invalid ESCO codes discards 110,573 train rows, 13,622 validation rows, and 13,587 test rows before trajectory-level filtering.

using Low-Rank Adaptation (LoRA) under the same last-step next-occupation objective. Implementation hyperparameters are reported in §4.

We next incorporate supervised first-order career-transition information derived exclusively from the training trajectories. For every consecutive occupation pair $o_i \rightarrow o_j$, the training data provide a transition signal S_{Markov}.

Retriever and transition scores are combined as shown in Eq. (1):

$$S_{\text{final}} = \alpha_{\text{Markov}} S_{\text{Markov}} + \alpha_{\text{model}} S_{\text{retriever}}. \tag{1}$$

Interpolation coefficients are selected on validation data from $\{0.0, 0.1, 0.25, 0.5, 0.75, 0.9, 1.0\}$.

The selected coefficients are frozen for test evaluation. Since retriever and transition scores may have different scales, the coefficient values are not interpreted as direct measures of the intrinsic importance of each signal.

The Markov component used here and the transition knowledge graph (Transition KG) introduced below are both derived from train-only transition statistics. They differ in scoring rather than data provenance. The Markov component uses a simple first-order transition score, whereas the Transition KG combines train-derived transition evidence with recent-history weighting, recency decay, exact and prefix evidence, and prior smoothing.

3.5 Frozen Occupational Wiki

To investigate whether generic occupational knowledge adds information beyond historical transitions, we construct a frozen Wiki containing one structured page for each of the 2,982 ESCO occupations.

Qwen3.5-9B [11] is used offline to generate these pages. Each page is generated only from the corresponding ESCO identifier, title, and description. The generated fields contain a summary, core tasks, skills or occupational context, and career-transition hints. No external documents are retrieved during generation.

All 2,982 pages are generated successfully, with no generation failures. Once constructed, the Wiki remains fixed throughout evaluation and receives no query-time modification or writeback.

Wiki-only retrieval uses Best Matching 25 (BM25) [13] over the frozen occupational pages. BM25 provides a deterministic textual interface to the generated pages without introducing another learned retriever.

3.6 Transition Knowledge Graph

We construct the Transition KG from the 256,593 training trajectories. Each occupation is represented as a node, and each observed consecutive transition contributes a directed edge $o_i \rightarrow o_j$.

The Transition KG scorer uses the five most recent occupations in the observed history, a recency decay of 0.65, an exact-transition weight of 8.0, a unit rank-bonus term, and a target-prior weight of 0.05. These Transition-KG hyperparameters were fixed heuristically during development before test evaluation and were not selected or tuned on the test split. The candidate pool is capped at 100 occupations.

The deterministic Wiki evaluation considers four configurations:

- **Wiki only**: retrieval from frozen occupational pages;
- **Transition KG**: transition-based candidate ranking;
- **Wiki+KG fallback**: KG candidates followed by Wiki/raw fallback when additional candidates are required;
- **Wiki+KG score fusion**: direct score-level fusion of Wiki and transition evidence.

For score fusion, Wiki and KG score lists are independently min–max normalized and combined as shown in Eq. (2):

$$S_{\text{Wiki+KG}} = \alpha S_{\text{KG}} + (1 - \alpha) S_{\text{Wiki}}. \tag{2}$$

The Wiki/KG fusion weight is selected from the validation-side alpha grid $\{0.7, 0.8, 0.9, 0.95\}$. The selected configuration uses $\alpha = 0.8$, corresponding to

$$S_{\text{Wiki+KG}} = 0.8 S_{\text{KG}} + 0.2 S_{\text{Wiki}}.$$

The resulting ranking is supplemented by raw ESCO fallback where required to complete the candidate pool.

3.7 Query-Time LLM Reasoning

Query-time Qwen reasoning is evaluated as an exploratory ablation rather than as part of the full-scale deterministic comparison.

For *direct selection*, deterministic retrieval first generates the top 100 occupation candidates. Qwen3.5-9B reads this list and directly selects the three most plausible candidate IDs. These selected occupations are moved to the beginning of the final ranking, while all remaining candidates preserve their original retrieval order.

For the *routing pipeline*, Qwen selects up to five Wiki pages from an 80-page routing shortlist before downstream candidate ranking. Because the pipeline contains both query-time routing and final LLM-based ranking, we treat it as a complete query-time LLM pipeline ablation rather than attributing every downstream change exclusively to routing.

Both LLM experiments operate on the same deterministic top-100 candidate pool. Qwen therefore changes ranking or selection behavior but does not alter the upstream candidate set.

For failed ranking calls in the routing experiment, the instance remains in the evaluation and the system falls back to the deterministic candidate ranking. No failed instance is removed from the metric denominator.

4 Experimental Setup

Metrics. We report **Accuracy at 1 (Acc@1)**, **Mean Reciprocal Rank at 10 (MRR@10)**, **Normalized Discounted Cumulative Gain at 10 (NDCG@10)**, and **Recall@10**. Candidate-generation experiments additionally report **Candidate Recall@100**. We omit **Mean Average Precision at 10 (MAP@10)** because each query contains a single relevant next occupation, making MAP@10 numerically identical to MRR@10.

Evaluation protocol. Validation data are used for model selection and interpolation-weight selection. The main deterministic retrieval and transition results are reported on the full 32,089-instance test split.

The deterministic Wiki/KG experiments are run on the complete validation and test splits, with test results reported in the main paper and validation results reported in the appendix.

The two query-time Qwen ablations use a fixed 2,000-instance validation subset with seed 42. This design keeps the test split untouched by exploratory LLM decisions and limits the inference cost of query-time generation.

Models. We evaluate BGE-M3 [1], Sentence Transformer [12], Qwen3-Embedding-8B [23], and LateOn [17]. Qwen3.5-9B [11] is used for frozen Wiki construction and the query-time LLM ablations. We select it because its open weights allow local deployment, while its 9B-parameter size provides a practical balance between model capacity and the inference cost required for thousands of query-time evaluations. The model is served through vLLM [9].

Task-adapted retrieval. For the Sentence Transformer, we initialize from `BAAI/bge-base-en-v1.5` and optimize with Multiple Negatives Ranking Loss for one epoch, using a batch size of 32, learning rate 2×10^{-5}, and four negatives per training instance.

For Qwen3-Embedding-8B, we initialize from Qwen/Qwen3-Embedding-8B and perform Low-Rank Adaptation (LoRA) with a per-device batch size of 2, gradient accumulation over 8 steps, one training epoch, learning rate 6×10^{-6}, seven negatives per instance, and a maximum input length of 512 tokens. Training uses 4-bit bitsandbytes quantization with bfloat16 computation.

Training examples follow the same last-step prediction protocol as evaluation, with negative sampling from the train-derived ESCO candidate pool.

Query-time LLM configuration. The direct-selection experiment uses temperature 0, top-p 1, a maximum output length of 128 tokens, structured output, and hard Top-3 reordering.

The routing pipeline uses deterministic decoding, a maximum of 128 output tokens for both routing and ranking, and a ranking bonus of 0.2. The routing run contains 265 ranking failures among 2,000 instances (13.3%). These instances remain in the evaluation and fall back to deterministic ranking.

Statistical analysis. We use paired bootstrap resampling with 10,000 samples and seed 42 to compare systems on the same evaluation instances. We report 95% bootstrap confidence intervals for the main paired comparisons.

Implementation details. Formal retrieval evaluation is performed on NVIDIA GeForce RTX 4090 GPUs with Python 3.10.12, PyTorch 2.7.1+cu128, and CUDA 12.8. The final Sentence Transformer evaluation uses `sentence-transformers` 5.7.0 and `transformers` 4.57.6, whereas the BGE-M3, Qwen3-Embedding-8B, and LateOn evaluation environments use `sentence-transformers` 5.1.1 and `transformers` 4.52.3; LateOn additionally uses PyLate 1.3.4.

These version differences reflect model-specific inference environments rather than different evaluation protocols: all systems are evaluated on the same queries and candidate vocabulary, and their saved rankings are scored with the same metric implementation.

Sentence Transformer, Qwen3-Embedding-8B, and BGE-M3 are evaluated with batch sizes of 64, 1, and 32, respectively. LateOn uses document and query batch sizes of 32 and 16.

For Qwen3-Embedding-8B retrieval, queries are prefixed with the instruction "Instruct: Given a career history, retrieve the most plausible next ESCO occupation."

The final Qwen3.5-9B Wiki-generation and query-time inference runs are served with vLLM `0.19.0` on an NVIDIA GeForce RTX 4090 GPU.

5 Results

5.1 Retriever-Only Ranking

Table 2 addresses RQ1. Among the standalone systems, the task-adapted Qwen3-Embedding-8B model achieves the highest MRR@10, NDCG@10, and Recall@10, whereas the task-adapted Sentence Transformer achieves the highest Acc@1.

BGE-M3 and LateOn are evaluated without task-specific retriever training and are therefore substantially weaker than the two adapted retrievers in this standalone comparison. The result also highlights why retriever-only and transition-aware results must be compared on the same split rather than across validation and test settings.

5.2 Transition-Aware Hybrid Ranking

Table 3 addresses RQ2. The trained Sentence Transformer becomes the strongest hybrid configuration, reaching an MRR@10 of 0.155 and Recall@10 of 0.340, followed closely by the trained Qwen3-Embedding-8B model.

For the trained Sentence Transformer, transition fusion increases MRR@10 from 0.111 to 0.155. The paired improvement is +0.0439, with a 95% bootstrap confidence interval of $[0.0409, 0.0467]$.

Compared with BGE-M3+Markov, the trained Sentence Transformer+Markov system improves MRR@10 by +0.0150, with a 95% confidence interval of $[0.0136, 0.0165]$. This indicates that task adaptation still provides a measurable benefit even after both systems receive transition information.

Retriever	Training	Acc@1	MRR@10	NDCG@10	Recall@10
BGE-M3	General-purpose	0.034	0.069	0.089	0.155
Sentence Transformer	Task-adapted	**0.053**	0.111	0.149	0.275
Qwen3-Embedding-8B	Task-adapted	0.047	**0.112**	**0.155**	**0.295**
LateOn	General-purpose	0.037	0.075	0.096	0.165

Table 2: Retriever-only performance on the 32,089-instance test split. No transition fusion or query-time LLM reasoning is used. Bold indicates the best value in each metric column.

Model	α_{Markov}	α_{Model}	Acc@1	MRR@10	NDCG@10	Recall@10
Sentence Transformer trained	0.25	0.75	**0.086**	**0.155**	**0.198**	**0.340**
Qwen3-Embedding-8B trained	0.25	0.75	0.085	0.152	0.195	0.336
BGE-M3	0.50	0.50	0.078	0.140	0.179	0.307
LateOn	0.90	0.10	0.073	0.133	0.172	0.299

Table 3: Final test performance with first-order transition fusion. Interpolation coefficients are selected on validation data and frozen for the 32,089-instance test split. Bold indicates the best value in each evaluation-metric column.

Both task-adapted dense retrievers favor a model-dominant configuration with $\alpha_{\text{model}} = 0.75$. BGE-M3 uses equal model and transition weights, while LateOn is strongly Markov-dominant with $\alpha_{\text{Markov}} = 0.90$. These coefficients should not be read as direct measures of signal importance because the component score distributions differ.

5.3 Frozen Wiki Knowledge versus Transition Statistics

Table 4 addresses RQ3. Wiki-only retrieval is substantially weaker than the Transition KG. Wiki-only achieves an MRR@10 of 0.056 and Candidate Recall@100 of 0.309, compared with 0.133 and 0.671 for the Transition KG. This indicates that empirical career-transition statistics provide substantially stronger predictive information than the frozen occupational Wiki text.

Fallback augmentation produces exactly the same ranking metrics as the Transition KG, showing that simply adding Wiki candidates does not improve prediction.

Score-level fusion produces a modest but statistically reliable complementary effect. Wiki+KG fusion improves MRR@10 from 0.133 to 0.138. The paired MRR@10 difference is +0.0047, with a 95% bootstrap confidence interval of $[0.0040, 0.0054]$. Recall@10 also increases from 0.317 to 0.324.

Interestingly, Candidate Recall@100 decreases from 0.671 for the Transition KG to 0.624 under score fusion despite improvements in the top-10 ranking metrics. The gain therefore arises primarily from reordering candidates near the top of the ranking rather than from expanding top-100 candidate coverage.

Overall, supervised transition statistics are the dominant signal, while frozen Wiki knowledge provides a modest complementary ranking benefit when combined through score-level fusion.

5.4 Query-Time LLM Reasoning

Table 5 addresses RQ4. Direct Top-3 selection clearly harms ranking quality. MRR@10 decreases from 0.134 to 0.102. The paired difference is -0.0315, with a 95% bootstrap confidence interval of $[-0.0405, -0.0227]$. The degradation is therefore statistically reliable rather than attributable to sampling noise.

The routing pipeline behaves differently. MRR@10 remains approximately unchanged at 0.134. The paired difference is +0.0005, with a 95% confidence interval of $[-0.0008, 0.0018]$. Because the interval contains zero, we find no reliable evidence of an improvement.

The routing pipeline also introduces 265 ranking failures, corresponding to 13.3% of the evaluation sample. These failed ranking calls remain in the 2,000-instance denominator and fall back to deterministic candidate ranking.

Moreover, the gold occupation occurs among the Wiki pages selected by the router in only 3.95% of instances. Candidate Recall@100 remains identical across the three settings because Qwen operates downstream of the same fixed top-100 candidate pool.

5.5 Error Analysis

We further examine where transition and semantic signals are most useful. Table 6 reports MRR@10 for return versus novel moves and for seen versus unseen final transitions.

Return moves are substantially easier for all systems. The effect is particularly strong for pure semantic retrieval: the task-adapted Sentence Transformer reaches 0.482 MRR@10 on return moves but only 0.046 on novel moves. Transition-aware systems substantially improve the much larger novel-move subset.

Seen and unseen transitions show a different pattern. Sentence Transformer+Markov is strongest when the final transition has been observed in training, reaching 0.224 MRR@10. For unseen transitions, however, the pure Sentence Transformer reaches 0.055, compared with 0.028 after Markov fusion. This indicates that direct

Method	Cand. R@100	Acc@1	MRR@10	NDCG@10	Recall@10
Wiki only	0.309	0.030	0.056	0.073	0.129
Transition KG	**0.671**	0.070	0.133	0.176	0.317
Wiki+KG fallback	**0.671**	0.070	0.133	0.176	0.317
Wiki+KG fusion	0.624	**0.073**	**0.138**	**0.181**	**0.324**

Table 4: Full-scale deterministic Wiki/KG comparison on the test split. Score fusion combines independently normalized KG and Wiki scores as $0.8S_{\text{KG}} + 0.2S_{\text{Wiki}}$. Bold indicates the best value in each metric column; tied best values are both highlighted.

Ablation	Setting	Cand. R@100	MRR@10	NDCG@10	Recall@10	Failures	s/instance	ΔMRR@10 (95% CI)
Direct selection	Deterministic baseline	0.690	0.134	0.176	0.314	0	–	–
Direct selection	+ Qwen Top-3	0.690	0.102	0.149	0.305	0	0.684	−0.0315 [−0.0405, −0.0227]
Routing pipeline	Deterministic baseline	0.690	0.134	0.176	0.314	0	–	–
Routing pipeline	+ Query-time Qwen	0.690	0.134	0.175	0.308	265 (13.3%)	1.170	+0.0005 [−0.0008, 0.0018]

Table 5: Query-time Qwen ablations on 2,000 validation instances. All settings operate on the same deterministic top-100 candidate pool; therefore Candidate Recall@100 is unchanged. Confidence intervals are obtained with paired bootstrap resampling.

Method	Return	Novel	Seen transition	Unseen transition
BGE-M3	0.382	0.014	0.085	0.038
BGE-M3+Markov	0.355	0.102	0.208	0.015
Sentence Transformer	**0.482**	0.046	0.141	**0.055**
Sentence Transformer+Markov	0.422	**0.108**	**0.224**	0.028
Transition KG	0.329	0.099	0.195	0.019
Wiki+KG fusion	0.385	0.095	0.200	0.024

Table 6: MRR@10 error analysis on the test split. Return moves contain 4,778 instances and novel moves 27,311 instances. Seen final transitions contain 20,764 instances and unseen transitions 11,325 instances. Bold indicates the best value in each subset column.

transition evidence is highly useful when it exists, while semantic representations are relatively more robust when the specific transition has not been observed.

Wiki+KG fusion improves the Transition KG for both seen and unseen transitions, with a larger Recall@10 benefit on unseen transitions. Additional analysis by career-history length and target frequency is reported in the appendix.

5.6 Worked Example

To make the ranking behavior concrete, we inspect a real non-return test instance whose observed career history is *administrative assistant* → *advertising assistant*, while the gold next occupation is *management assistant*. The gold occupation does not appear in the observed career history.

Table 7 shows a non-return instance in which Wiki evidence changes the top of the ranking. The Transition KG follows train-derived transition evidence from the recent career history and places the gold occupation at rank 11. Wiki-only retrieval produces a ranking that is more closely aligned with lexical and occupational similarity among administrative-support roles, placing *management assistant* at rank 4. After score fusion, the gold occupation moves to rank 2, while the highest ranked candidates remain plausible administrative-support occupations. This example illustrates how frozen Wiki evidence can complement transition statistics without replacing them.

5.7 Discussion

Three patterns emerge from our experiments.

Transition structure is a strong task-specific signal. The standalone retrieval comparison shows that task-adapted semantic encoders provide useful occupation rankings, but the transition-aware experiments demonstrate that next-occupation prediction cannot be reduced to semantic similarity alone. Adding train-derived transition information substantially improves the Sentence Transformer, and the Transition KG alone reaches an MRR@10 of 0.133.

Frozen occupational knowledge is complementary, not primary. Wiki-only retrieval is substantially weaker than the Transition KG. However, score-level Wiki+KG fusion provides a small but statistically reliable top-10 improvement. Generic occupational knowledge therefore appears most useful as an auxiliary ranking signal rather than as a replacement for supervised transition statistics.

Method	Top-1	Top-2	Top-3	Gold rank
Transition KG	administrative assistant	sales support assistant	public relations officer	11
Wiki only	advertising assistant	administrative assistant	legal administrative assistant	4
Wiki+KG fusion	administrative assistant	management assistant	sales support assistant	2

Table 7: Worked example for a non-return test instance. The observed career history is *administrative assistant* → *advertising assistant*, and the gold next occupation is *management assistant*.

Query-time LLM reasoning does not guarantee improvement. Direct candidate selection significantly reduces ranking quality, while the routing pipeline adds latency and ranking failures for a change that is statistically indistinguishable from zero. In this setting, deterministic transition modeling and offline knowledge fusion provide a better accuracy–complexity trade-off than additional query-time LLM reasoning.

Calibration against STEP.. STEP [7] reports an MRR of 22.7% and Recall@10 of 41.1% for its full JobHop v2 model, while its Markov baseline reaches Recall@10 of 34.2%. Our Transition KG reaches Recall@10 of 31.7%, which is broadly comparable in scale to the STEP Markov result.

The MRR values should not be compared directly because STEP reports full MRR, whereas we report MRR@10. The two pipelines may also differ in preprocessing, candidate construction, and feature representation. Nevertheless, the similar Recall@10 performance of the transition baselines suggests that the two evaluation protocols are broadly comparable in scale.

6 Conclusion

We presented a diagnostic study of semantic retrieval, supervised transition statistics, frozen occupational knowledge, and query-time LLM reasoning for next-occupation prediction.

Across the experiments, train-derived career-transition structure emerges as a particularly strong task-specific source of evidence. Task-adapted semantic retrieval remains useful, especially for instances in which the specific transition has not been observed, but transition-aware ranking provides substantial gains on the aggregate task.

The frozen Wiki experiments further clarify the role of occupational knowledge. Wiki-only retrieval is substantially weaker than the Transition KG, while score-level Wiki+KG fusion provides a small but statistically reliable complementary improvement.

Finally, query-time Qwen reasoning does not reliably improve the deterministic pipeline. Direct Top-3 selection significantly reduces ranking quality, while the routing pipeline introduces latency and ranking failures without a measurable MRR improvement.

Overall, our findings suggest that strong transition modeling should be established before introducing more expensive generative inference. Future work should investigate ESCO hierarchy-aware ranking, rare and unseen career transitions, cross-dataset evaluation, and richer agentic retrieval strategies.

Limitations. Our study focuses on a single career-trajectory dataset and a fixed ESCO occupation vocabulary, so the conclusions may not transfer directly to other labor markets, taxonomies, or vacancy-level recommendation settings.

Historical transition statistics reflect patterns present in the dataset and should not be interpreted as universally desirable career pathways. The query-time LLM experiments are evaluated on 2,000 validation instances rather than the complete split because of inference cost and because they are used as exploratory design ablations.

Finally, the query-time experiments evaluate one open-weight LLM family and a limited set of routing and candidate-selection strategies. The negative result should therefore not be interpreted as evidence that all forms of LLM-based reasoning are ineffective for next-occupation prediction.

Ethical considerations. Historical career trajectories may reflect structural inequalities and occupational segregation. A system trained to reproduce common transitions can therefore reinforce historical patterns rather than identify individually desirable opportunities.

The models evaluated in this work are research systems for next-occupation ranking. Their predictions should not be interpreted as normative career advice or used directly in automated hiring decisions without additional fairness, temporal-shift, and user-centered evaluation.

GenAI Usage Disclosure

Generative AI (GenAI) tools were utilized in the preparation of this manuscript to assist with coding and language polishing. The core research ideas, experimental design, analysis, and the overall drafting and writing of the paper were conducted by the authors without the involvement of GenAI.

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A Frozen Wiki Generation

The frozen occupational Wiki contains one structured entity page for every ESCO occupation in the candidate vocabulary. Each page stores the following fields: `docid`, `title`, `summary`, `core_tasks`, `skills_or_context`, and `career_transition_hints`.

Wiki construction produces 2,982 pages from 2,982 input occupations with zero generation failures. The complete construction takes 4,622.78 seconds, corresponding to approximately 1.55 seconds per occupation. The resulting pages are frozen before all downstream evaluation.

The Wiki-generation system message is:

```
Return JSON only. You write concise wiki
entity pages for ESCO occupations.
```

The user prompt template is:

```
Create a concise LLM-Wiki entity page for
this ESCO occupation.

Source occupation:
id: <ESCO_ID>
title: <TITLE>
description: <DESCRIPTION>

Return JSON only with this schema:
{
  "docid": "<ESCO_ID>",
  "title": "...",
  "summary": "...",
  "core_tasks": ["...", "..."],
  "skills_or_context": ["...", "..."],
  "career_transition_hints": ["...", "..."]
}

Rules:
- Do not invent another ESCO id.
- Use only information supported by the
  source occupation.
- Keep each field concise.
```

B Qwen Direct-Selection Prompt

For the direct-selection ablation, each candidate is represented as

```
id=<ESCO_ID> | title=<TITLE> |
```

```
source=<SOURCE> | summary=<SUMMARY>
```

where the summary is truncated to 60 characters in the final run. The system message is:

```
Return JSON only. Directly select the most
plausible ESCO occupation IDs from the
provided candidate list.
```

The user prompt template is:

```
Select the most plausible ESCO occupation IDs
for JobHop next-occupation prediction.

Career history query:
<CAREER_HISTORY_QUERY>

Candidate occupation IDs:
<CANDIDATE_TABLE>

Return JSON only:
{
  "selected_ids": ["id1", "id2"]
}

Rules:
- Select ONLY ESCO ids from the Candidate
  occupation IDs list.
- ESCO ids look like "2432.9" or "1221.3.2";
  do not output row numbers such as "1",
  "2", or "47".
- Return up to 3 ids in ranked order.
- Do NOT score candidates.
- Do NOT explain candidates.
- Do NOT output occupations outside the
  candidate list.
- Prefer plausible next occupations, not
  occupations already in the career history.
- If uncertain, preserve the original
  candidate order.
```

The final run uses a candidate pool of 100, temperature 0, top-p 1, a maximum output length of 128 tokens, concurrency 12, and hard Top-3 reordering.

C Qwen Routing and Ranking Prompts

The routing ablation first applies BM25 to obtain an 80-page Wiki shortlist. Qwen3.5-9B then selects at most five pages and produces up to four auxiliary search terms.

The routing system message is:

```
Return JSON only. You route JobHop queries
to ESCO Wiki pages.
```

The routing prompt template is:

```
Route this JobHop query to relevant ESCO
Wiki entity pages.

Career history query:
<CAREER_HISTORY_QUERY>

Candidate Wiki index:
<WIKI_INDEX>

Return JSON only:
{
  "selected_page_ids": ["id1", "id2"],
  "search_terms": ["term1", "term2"]
}

Rules:
- Select at most 5 page ids.
- Use only page ids from the Candidate
  Wiki index.
- Search terms should help grep the frozen
  Wiki for additional relevant pages.
- Prefer plausible next occupations, not
  past occupations from the query.
```

After candidate generation, the same model performs a final ranking step. Two of the highest-ranked candidate Wiki pages are included as detailed context, while all 100 candidates are presented in compact form.

The ranking system message is:

```
Return JSON only. You rank retrieved ESCO
candidate IDs for JobHop.
```

The ranking prompt template is:

```
Rank candidate ESCO occupation IDs for
JobHop next-occupation prediction.

Career history query:
<CAREER_HISTORY_QUERY>

Retrieved Wiki context:
<WIKI_CONTEXT>

Candidate occupation IDs:
<CANDIDATE_TABLE>

Return JSON only:
{
  "ranked_ids": ["id1", "id2"]
}

Rules:
- Rank ONLY ESCO ids from the Candidate
  occupation IDs list.
- ESCO ids look like "2432.9" or
  "1221.3.2"; do not output row numbers.
- Return up to 10 ids.
- Prefer plausible next occupations, not
  occupations already in the career history.
- If uncertain, preserve the candidate order.
```

The final routing run uses temperature 0, top-p 1, 128-token limits for both routing and ranking, concurrency 8, and a ranking bonus of 0.2. The Qwen ranking is incorporated as a bonus over the deterministic candidate ranking rather than replacing it completely.

D Additional Error Analysis

History	N	KG	Wiki+KG	ST	ST+M
1	4,849	0.116	0.113	0.057	0.127
2–3	11,342	0.138	0.142	0.114	0.155
4–7	12,685	0.135	0.142	0.121	0.161
8–15	3,177	0.135	0.146	0.143	0.171
16+	36	0.157	0.186	0.174	0.201

Table 8: MRR@10 by observed career-history length. KG denotes the Transition KG, ST denotes Sentence Transformer, and ST+M denotes Sentence Transformer+Markov. The 16+ bucket contains only 36 instances and should not be over-interpreted.

Frequency	N	KG	Wiki+KG	ST	ST+M
100+	25,479	0.166	0.171	0.129	0.188
21–100	5,075	0.007	0.010	0.047	0.030
6–20	1,246	0.003	0.005	0.017	0.011
1–5	269	0.000	0.001	0.016	0.007
0	20	0.000	0.000	0.000	0.000

Table 9: MRR@10 by target occupation frequency in the training trajectories. KG denotes the Transition KG, ST denotes Sentence Transformer, and ST+M denotes Sentence Transformer+Markov. The buckets provide a head-to-tail view of target occupations.

E Validation Wiki/KG Results

Method	C.R@100	A@1	MRR	NDCG	R@10
Wiki only	0.318	0.030	0.058	0.076	0.136
Transition KG	**0.676**	0.070	0.133	0.176	0.317
Wiki+KG fallback	**0.676**	0.070	0.133	0.176	0.317
Wiki+KG fusion	0.626	**0.072**	**0.139**	**0.182**	**0.326**

Table 10: Validation Wiki/KG results. Reported separately from the test results used for the main comparison. Bold indicates the best value in each metric column; tied best values are both highlighted.

F Additional Implementation Details

Data preparation. All retriever experiments use the same JobHop preprocessing pipeline with random seed 7. Unknown occupation codes are removed, jobs are sorted chronologically, and consecutive duplicate occupations are collapsed. Only trajectories containing at least two valid occupations after duplicate collapse are retained.

The candidate vocabulary consists only of ESCO occupations observed in the retained training trajectories. Validation and test targets are checked against this train-derived candidate vocabulary. In the final preprocessing run, this check removes zero validation or test instances.

Training examples follow the last-step prediction protocol used at evaluation time. Negative occupations for retriever training are sampled randomly from the training candidate vocabulary.

Retriever initialization. The task-adapted Sentence Transformer is initialized from `BAAI/bge-base-en-v1.5`, while the task-adapted Qwen retriever is initialized from `Qwen/Qwen3-Embedding-8B`. The general-purpose comparison additionally evaluates `BAAI/bge-m3` and `lightonai/LateOn`.

Retrieval environment. Formal retrieval evaluation runs on NVIDIA GeForce RTX 4090 GPUs with Python 3.10.12, PyTorch 2.7.1+cu128, and CUDA 12.8. The Sentence Transformer final evaluation uses `sentence-transformers` 5.7.0 and `transformers` 4.57.6. The BGE-M3, Qwen3-Embedding-8B, and LateOn formal evaluations use `sentence-transformers` 5.1.1 and `transformers` 4.52.3, with PyLate 1.3.4 for LateOn.

Transition candidate generation. Both the Markov component and the Transition KG are constructed exclusively from occupation transitions observed in the training trajectories. The Markov component uses a simple first-order transition score. The Transition KG extends the same transition source with history-aware scoring.

Candidate generation uses the five most recent occupations, recency decay 0.65, exact-transition weight 8.0, a unit rank-bonus term, and target-prior weight 0.05. These Transition-KG hyperparameters were fixed heuristically during development before test evaluation and were not selected or tuned on the test split. The candidate pool is capped at 100 occupations.

Wiki score fusion. Wiki and KG scores are independently min–max normalized. The Wiki/KG fusion weight is selected from the validation-side alpha grid $\{0.7, 0.8, 0.9, 0.95\}$. The selected score-level fusion configuration uses $0.8S_{\text{KG}} + 0.2S_{\text{Wiki}}$, followed by raw ESCO fallback when required to complete the candidate pool.

Query-time Qwen ablations. Both Qwen3.5-9B ablations use a fixed sample of 2,000 validation instances with seed 42. Direct selection uses a 100-candidate input, hard Top-3 reordering, temperature 0, top-p 1, a 128-token output limit, and concurrency 12.

Routing uses an 80-page initial Wiki shortlist, selects up to five pages, uses a 100-candidate final pool, returns up to ten ranked IDs, and applies a ranking bonus of 0.2. Its routing and ranking outputs are each limited to 128 tokens, with concurrency 8.

The routing run contains 265 ranking failures. Failed ranking calls remain in the 2,000-instance evaluation and use the deterministic candidate ranking as fallback. Candidate Recall@100 is unchanged because both LLM ablations operate downstream of the same fixed top-100 candidate set.

The final Qwen3.5-9B inference runs are served using vLLM 0.19.0's OpenAI-compatible API server on an NVIDIA GeForce RTX 4090 GPU. The server uses a maximum model length of 4096, GPU memory utilization of 0.90, eager execution, and trusted remote model code.